PSTAT100_Lab3_extension

Scenario 2: hawks

In this scenario, you'll explore sampling from a population with group structure – frequently bias can arise from inadvertent uneven sampling of groups within a population.

Hypothetical population

Suppose you're interested in determining the average beak-to-tail length of red-tailed hawks to help differentiate them from other hawks by sight at a distance. Females and males differ slightly in length – females are generally larger than males. The cell below generates length measurements for a hypothetical population of 3000 females and 2000 males.

```
# Load libraries
library(dplyr)
library(ggplot2)
```

```
# Set seed for reproducibility
set.seed(40721)

# Simulate hypothetical population for female hawks
female_hawks <- data.frame(
  length = rnorm(n = 3000, mean = 57.5, sd = 3),
  sex = rep("female", 3000)
)

# Simulate hypothetical population for male hawks
male_hawks <- data.frame(
  length = rnorm(n = 2000, mean = 50.5, sd = 3),
  sex = rep("male", 2000)
)
```

```r
# Combine the two populations
population_hawks <- rbind(female_hawks, male_hawks)

# Preview the first 2 rows for each sex
population_hawks[population_hawks$sex == "female", ][1:2, ]
```

```
    length    sex
1 60.21225 female
2 57.44555 female
```

```r
population_hawks[population_hawks$sex == "male", ][1:2, ]
```

```
       length  sex
3001 45.23748 male
3002 47.32008 male
```

The cell below produces a histogram of the lengths in the population overall (bottom panel) and when distinguished by sex (top panel).

```r
# Load required libraries
library(ggplot2)
library(patchwork)

# Base plot for population_hawks
base <- ggplot(population_hawks, aes(x = length))

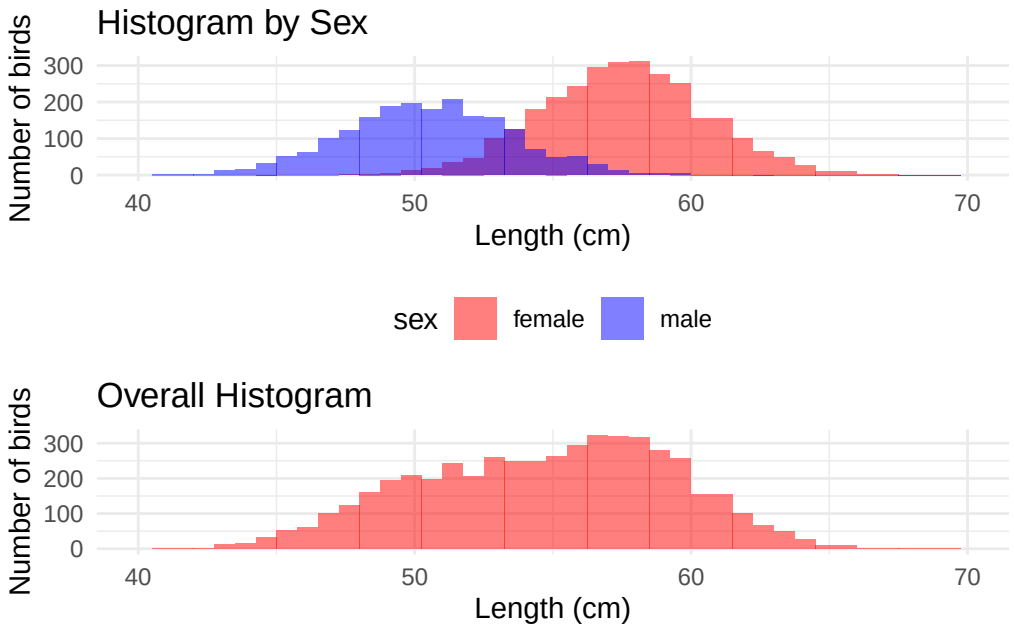
# Histogram for the overall population
hist <- base +
  geom_histogram(binwidth = 0.75, fill = "red", alpha = 0.5, boundary = 0) +
  scale_x_continuous(limits = c(40, 70)) +
  labs(
    x = "Length (cm)",
    y = "Number of birds",
    title = "Overall Histogram"
  ) +
  theme_minimal() +
  theme(plot.margin = margin(5, 5, 5, 5))

# Histogram grouped by sex
hist_bysex <- base +
```

```
  geom_histogram(aes(fill = sex), binwidth = 0.75, alpha = 0.5, position = "identity", bounda
  scale_x_continuous(limits = c(40, 70)) +
  scale_fill_manual(values = c("red", "blue")) +
  labs(
    x = "Length (cm)",
    y = "Number of birds",
    title = "Histogram by Sex"
  ) +
  theme_minimal() +
  theme(plot.margin = margin(5, 5, 5, 5), legend.position = "bottom")

# Combine the histograms vertically
combined_plot <- hist_bysex / hist

# Display the combined plot
print(combined_plot)
```



The population mean – average length of both female and male red-tailed hawks – is shown below.

```
# Compute the mean for numeric columns only
col_means <- sapply(population_hawks[sapply(population_hawks, is.numeric)], mean)
```

```
# Display the results
col_means
```

```
  length
54.69896
```

Question 8

First try drawing a random sample from the population with seed 40821:

YOUR ANSWER:

```
# Set seed for reproducibility
set.seed(40821)

# Randomly sample 300 rows without replacement
sample_hawks <- population_hawks[sample(1:nrow(population_hawks), size = 300, replace = FAI

# View the sampled data
head(sample_hawks)
```

```
       length    sex
3077 49.97600   male
4630 49.55513   male
4763 53.43223   male
1272 61.47602 female
1718 55.31940 female
1175 60.80274 female
```

Do you expect that the sample will contain equal numbers of male and female hawks? Think about this for a moment (you don't have to provide a written answer), and then compute the proportions of individuals in the sample of each sex and store the result as a dataframe named `proportion_hawks_sample`. The dataframe should have one column named `proportion` and two rows indexed by `sex`.

Hint: group by sex, use `.count()`, and divide by the sample size. Be sure to rename the output column appropriately, as the default behavior produces a column called `length`.

YOUR ANSWER:

```r
# Calculate the proportions of each sex in the sample
proportion_hawks_sample <- sample_hawks %>%
  group_by(sex) %>%
  summarise(proportion = n() / nrow(sample_hawks))

# Display the proportions
proportion_hawks_sample
```

```
# A tibble: 2 x 2
  sex    proportion
  <chr>       <dbl>
1 female      0.623
2 male        0.377
```

The sample mean is shown below, and is fairly close to the population mean. This should be expected, since you already saw in scenario 1 that random sampling is an unbiased sampling design with respect to the mean.

YOUR ANSWER:

```r
# Compute the mean for numeric columns only
sample_hawks_means <- sapply(sample_hawks[sapply(sample_hawks, is.numeric)], mean)

# Display the results
sample_hawks_means
```

```
  length
54.96569
```

Biased sampling

Let's now consider a biased sampling design. Usually, length measurements are collected from dead specimens collected opportunistically. Imagine that male mortality is higher, so there are better chances of finding dead males than dead females. Suppose in particular that specimens are five times as likely to be male; to represent this situation, we'll assign sampling weights of 5/6 to all male hawks and weights of 1/6 to all female hawks.

```r
# Define the weight function
weight_fn <- function(sex, p = 5/6) {
  if (sex == "male") {
```

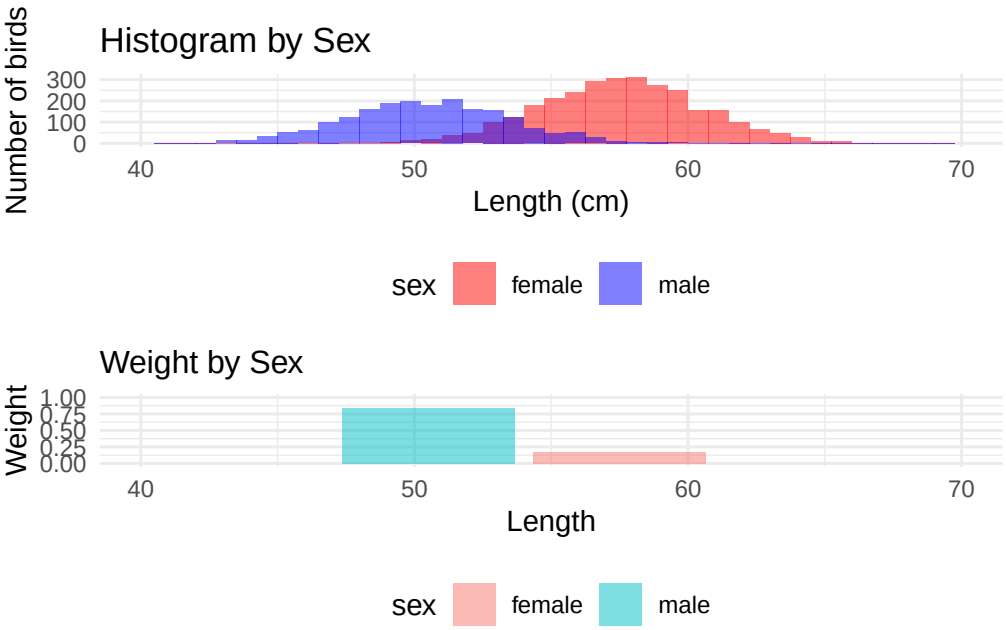
```
    out <- p
  } else {
    out <- 1 - p
  }
  return(out)
}

# Create the weight data frame
weight_df <- data.frame(
  length = c(50.5, 57.5),
  weight = c(5/6, 1/6),
  sex = c("male", "female")
)

# Plot the weights as a bar chart
wt <- ggplot(weight_df, aes(x = length, y = weight, fill = sex)) +
  geom_bar(stat = "identity", alpha = 0.5) +
  scale_x_continuous(limits = c(40, 70)) +
  scale_y_continuous(limits = c(0, 1)) +
  labs(
    x = "Length",
    y = "Weight",
    title = "Weight by Sex"
  ) +
  theme_minimal() +
  theme(
    legend.position = "bottom",
    plot.title = element_text(size = 12)
  )

# Combine the bar chart with hist_bysex (assumes hist_bysex is already defined)
combined_plot <- hist_bysex / wt

# Display the combined plot
print(combined_plot)
```



Question 9

Draw a weighted sample `sample_hawks_biased` from the population `population_hawks` using the weights defined by `weight_fn`, and compute and store the value of the sample mean as `sample_hawks_biased_mean`.

YOUR ANSWER:

```r
# For reproducibility
set.seed(40821)

# Define the weight function
weight_fn <- function(sex, p = 5/6) {
  if (sex == "male") {
    return(p)
  } else {
    return(1 - p)
  }
}

# Assign weights to the population
population_hawks$weight <- sapply(population_hawks$sex, weight_fn)

# Draw a weighted sample
sample_hawks_biased <- population_hawks[sample(1:nrow(population_hawks), size = 300, replac

# Compute the mean of the sample
sample_hawks_biased_mean <- mean(sample_hawks_biased$length)

# Display the result
sample_hawks_biased_mean
```

```
[1] 52.16694
```

Question 10

Investigate the bias of the sample mean by:

- drawing 1000 samples with observations weighted by `weight_fn`;
- storing the sample mean from each sample as `samp_means_hawks`;
- computing the average difference between the sample means and the population mean and storing the resulting value as `avg_diff_hawks`.

YOUR ANSWER:

```r
# For reproducibility
set.seed(40221)

# Number of samples to simulate
nsim <- 1000

# Storage for the sample means
samp_means_hawks <- numeric(nsim)

# Define the weight function
weight_fn <- function(sex, p = 5/6) {
  if (sex == "male") {
    return(p)
  } else {
    return(1 - p)
  }
}

# Assign weights to the population
population_hawks$weight <- sapply(population_hawks$sex, weight_fn)

# Repeatedly sample and store the sample mean in the samp_means array
for (i in 1:nsim) {
  sample_hawks <- population_hawks[
    sample(
      1:nrow(population_hawks),
      size = 300,
      replace = FALSE,
      prob = population_hawks$weight
    ), ]
  samp_means_hawks[i] <- mean(sample_hawks$length)
}

# Compute the average difference between the sample means and the population mean
avg_diff_hawks <- mean(samp_means_hawks - mean(population_hawks$length))

# Display the result
avg_diff_hawks
```

```
[1] -2.544425
```

Question 11

Reflect a moment on your simulation result in question 10. If instead *female* mortality is higher and specimens for measurement are collected opportunistically, as described in the previous sampling design, do you expect that the average length in the sample will be an underestimate or an overestimate of the population mean? Explain why in 1-2 sentences, and carry out a simulation to check your intuition.

YOUR ANSWER:

```r
# Set seed for reproducibility
set.seed(40221)

# Invert weights: Higher weights for females and lower for males
weight_fn_inverted <- function(sex, p = 5/6) {
  if (sex == "male") {
    return(1 - p) # Lower weight for males
  } else {
    return(p)     # Higher weight for females
  }
}

# Assign inverted weights to the population
population_hawks$weight_inverted <- sapply(population_hawks$sex, weight_fn_inverted)

# Number of samples to simulate
nsim <- 1000

# Storage for the sample means
samp_means_hawks_inverted <- numeric(nsim)

# Repeatedly sample and store the sample mean
for (i in 1:nsim) {
  sample_hawks_inverted <- population_hawks[
    sample(
      1:nrow(population_hawks),
      size = 300,
      replace = FALSE,
      prob = population_hawks$weight_inverted
    ), ]
  samp_means_hawks_inverted[i] <- mean(sample_hawks_inverted$length)
}

# Compute bias: Average difference between sample means and population mean
estimated_bias <- mean(samp_means_hawks_inverted - mean(population_hawks$length))

# Display the result
estimated_bias
```

```
[1] 1.954291
```

Answer: If female mortality is higher, more females will be included in the sample. Since

females have larger average lengths than males, this would cause the sample to overestimate the population mean because the sample will overrepresent females with longer average lengths.

Bias correction

What can be done if a sampling design is biased? Is there any remedy?

You've seen some examples above of how bias can arise from a sampling mechanism in which units have unequal chances of being selected in the sample. Ideally, we'd work with random samples all the time, but that's not very realistic in practice. Fortunately, biased sampling is not a hopeless case – **it is possible to apply bias corrections if you have good information about which individuals were more likely to be sampled**.

To illustrate how this would work, let's revisit the biased sampling in scenario 1 – sampling larger eucalyptus seeds more often than smaller ones. Imagine you realize the mistake and conduct a quick experiment with your sieve to determine the proportion of seeds of each size that pass through, and use this to estimate the inclusion probabilities. (To simplify this excercise, we'll just use sampling weights we defined to calculate the actual inclusion probabilities.)

The cell below generates the population and sample from scenario 2 again:

```
# Set seed for reproducibility
set.seed(40221)

# Simulate seed diameters
population3 <- data.frame(
  seed = 0:(5000 - 1),
  diameter = rgamma(n = 5000, shape = 2, scale = 1/2)
)

# Define the weight function for inclusion probability
weight_fn <- function(x, r = 2, c = 2) {
  1 / (1 + exp(-r * (x - c)))
}

# Assign inclusion probability to each seed
population3$samp_weight <- weight_fn(population3$diameter)

# Draw weighted sample
set.seed(40721) # For reproducibility
sample3 <- population3[sample(1:nrow(population3), size = 250, replace = FALSE, prob = popula
```

```r
# View the weighted sample
head(sample3)
```

```
     seed  diameter samp_weight
2810 2809 0.9705148  0.11314911
3342 3341 1.9714175  0.48571265
4609 4608 1.7186179  0.36290813
2110 2109 3.2546002  0.92478429
90     89 1.6047183  0.31204770
3713 3712 0.4314770  0.04160475
```

The sample mean and population mean you calculated earlier are shown below:

```r
# Calculate sample and population means
means <- c(
  `sample mean` = mean(sample3$diameter),
  `population mean` = mean(population3$diameter)
)

# Print the means
print(means)
```

```
    sample mean population mean
      1.7644934       0.9968275
```

We can obtain an unbiased estimate of the population mean by computing a *weighted average* of the diameter measurements instead of the sample average after weighting the measurements in inverse proportion to the sampling weights:

$$\text{weighted average} = \sum_{i=1}^{250} \underbrace{\left(\frac{w_i^{-1}}{\sum_{j=1}^{250} w_j^{-1}} \right)}_{\text{bias adjustment}} \times \text{diameter}_i$$

This might look a little complicated, but the idea is simple – the weighted average corrects for bias by simply up-weighting observations with a lower sampling weight and down-weighting observations with a higher sampling weight.

The cell below performs this calculation.

```r
sample3$bias_adjustment <- (sample3$samp_weight^(-1)) / sum(sample3$samp_weight^(-1))

# Weight diameter measurements
sample3$weighted_diameter <- sample3$diameter * sample3$bias_adjustment

# Sum to compute weighted average
weighted_average <- sum(sample3$weighted_diameter)

# Display the result
weighted_average
```

```
[1] 0.8933892
```

Notice that the weighted average successfully corrected for the bias:

```r
# Calculate sample mean, weighted average, and population mean
means <- c(
  `sample mean` = mean(sample3$diameter),
  `weighted average` = sum(sample3$weighted_diameter),
  `population mean` = mean(population3$diameter)
)

# Print the results
print(means)
```

```
     sample mean weighted average  population mean
       1.7644934        0.8933892        0.9968275
```

Question 12

Perform a bias correction for the hawk scenario, using the actual inclusion probabilities, in the following steps:

- redraw `sample_hawks_biased` from question 9 exactly (use the same RNG seed) and retain the `weight` column
- compute the 'raw' sample mean and store the resulting value as `sample_mean`
- compute a bias adjustment factor as illustrated above by normalizing the inverse weights, and append the adjustment factor as a new column in `sample_hawks_biased` named `bias_adjustment`
- compute a weighted average length and store the resulting value as as `sample_mean_corrected`

Print the raw and corrected sample means.

YOUR ANSWER:

```r
set.seed(40821) # for reproducibility

# Assuming hawks_biased is already available
# Sample with bias and retain weights
sample_hawks_biased <- population_hawks[sample(1:nrow(population_hawks), size = 100, replac

# Compute the raw sample mean
sample_mean <- mean(sample_hawks_biased$length)

# Define the bias correction factor by normalizing the inverse weights
total_weight <- sum(1 / sample_hawks_biased$weight)
sample_hawks_biased$bias_adjustment <- (1 / sample_hawks_biased$weight) / total_weight

# Compute the weighted average length
sample_mean_corrected <- sum(sample_hawks_biased$length * sample_hawks_biased$bias_adjustme

# Print the raw and corrected sample means
print(paste("Raw Sample Mean:", sample_mean))
```

```
[1] "Raw Sample Mean: 52.3417468362372"
```

```r
print(paste("Corrected Sample Mean:", sample_mean_corrected))
```

```
[1] "Corrected Sample Mean: 54.8445863770591"
```